

RETHINKING CLASSIFIER-FREE GUIDANCE IN ON-POLICY DIFFUSION DISTILLATION

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<https://rethinking-cfg-opd.github.io>

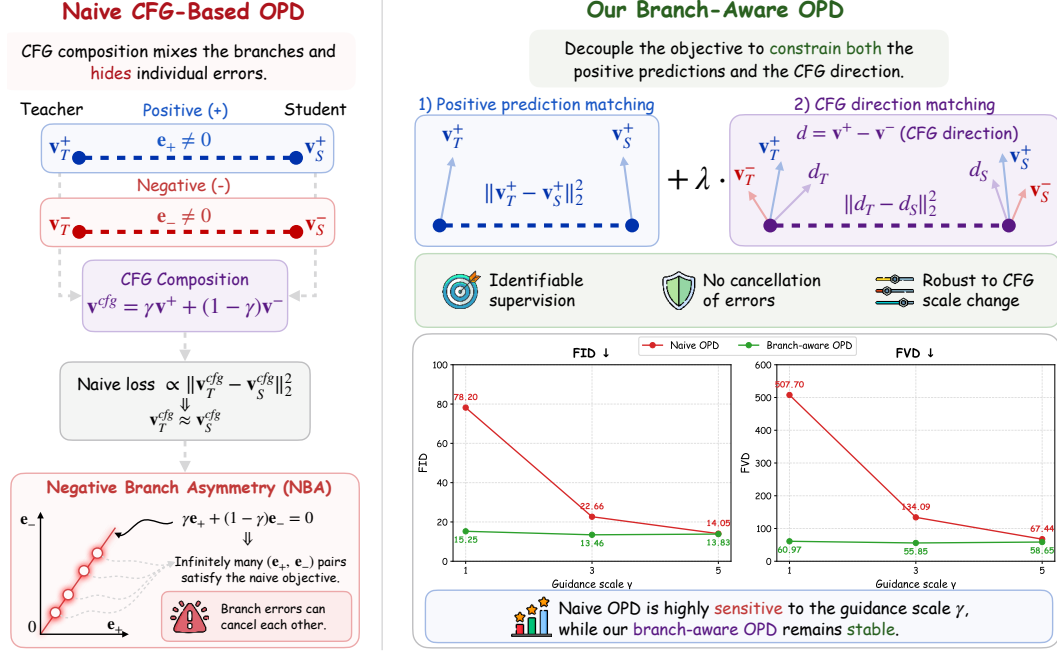


Figure 1: **Naive CFG-based OPD suffers from branch ambiguity.** Matching only the CFG-composed prediction allows positive and negative branch errors to cancel, creating infinitely many degenerate solutions. Our branch-aware objective, Positive-Direction Matching (PDM), resolves this ambiguity by separately constraining the positive prediction and the CFG conditional direction, restoring identifiable supervision and robust distillation across guidance scales.

ABSTRACT

On-policy distillation (OPD) adapts diffusion models by querying a teacher along trajectories generated by the current student, but how it should behave under classifier-free guidance (CFG), a default component of modern diffusion systems, remains poorly understood. Existing OPD methods naturally extend velocity matching to the CFG-composed prediction, matching teacher and student guided velocities directly. We show that this objective is valid in standard text-to-image settings, where teacher and student remain aligned on the negative branch, but can fail in more general settings with asymmetric teacher-student conditioning. In such cases, CFG composition couples positive- and negative- branch discrepancies, allowing branch-level errors to cancel while producing identical guided supervision signals. We term this phenomenon Negative Branch Asymmetry (NBA). To address NBA, we introduce Positive-Direction Matching (PDM), a branch-aware OPD objective that separately constrains the positive prediction and the CFG conditional direction. On dense-to-sparse video control, naive guided matching suffers from CFG-scale sensitivity under NBA while branch-aware supervision enables more robust and effective knowledge transfer; a FLUX-based

reference-conditioned visual-knowledge case study illustrates the same failure qualitatively.

1 INTRODUCTION

Classifier-free guidance (CFG) (Ho & Salimans, 2022) drives nearly every modern diffusion model (Labs, 2025; Esser et al., 2024; Wan et al., 2025), yet the velocity it denoises with is never produced by the network directly. At each step the model composes it from a conditional and an unconditional branch, $\tilde{\mathbf{v}} = \gamma \mathbf{v}^+ + (1 - \gamma) \mathbf{v}^-$, where the guidance scale γ is a deployment knob chosen at inference, often long after training. Any method that adapts a guided diffusion model must therefore decide what it is really supervising: the single composed velocity the model will deploy, or the two branch predictions that compose it.

On-policy distillation (OPD) adapts a diffusion model by letting the student generate a denoising trajectory with its current policy, evaluating teacher and student at the same student-visited states, and updating the student to match the teacher’s local velocity (Jiang et al., 2026; Fang et al., 2026; Li et al., 2026b). The combination of on-policy state coverage and dense per-step supervision makes OPD an appealing alternative to offline distillation and scalar-reward optimization (Agarwal et al., 2024; Zhong et al., 2026).

Faced with a guided teacher, diffusion distillation can handle CFG in different ways. Some methods absorb a CFG-defined teacher field into a single student prediction, thereby internalizing the guidance strength used to construct the target (Luo et al., 2023; Yin et al., 2024b; Zhou et al., 2026). In the setting considered here, however, both teacher and student retain the standard CFG interface of the underlying diffusion model and produce separate positive and negative predictions (Fang et al., 2026; Li et al., 2026b). A natural extension of OPD is then to match their CFG-composed velocities, $\mathcal{L}_{\text{naive}} \propto \|\tilde{\mathbf{v}}^T - \tilde{\mathbf{v}}^S\|_2^2$. Because the two branches remain available, the student can still be evaluated with different guidance scales at inference; however, the training loss constrains only their composition at the training scale.

Composed matching is well behaved in the text-to-image regime studied so far, where teacher and student are instantiated from the same base model and share an identical null-text negative branch. Their negative predictions are therefore expected to agree, causing composed matching to reduce to positive-branch matching and recover the standard OPD velocity-matching objective.

That expectation quietly fails the moment the teacher is given information the student lacks. Consider distilling reference-conditioned visual knowledge: a reference image supplies an appearance that text alone cannot pin down, such as the exact look of a specific character, and we want a text-only student to reproduce it from the prompt alone. The teacher sees the reference on *both* CFG branches, positive and negative, while the student sees only text on both. Teacher and student now differ not just on the positive branch but on the negative branch as well. Composed matching no longer has a clean target to reduce to.

When the branches disagree on both sides, matching only the composed velocity is under-identified. Let $\mathbf{e}_+ = \mathbf{v}_T^+ - \mathbf{v}_S^+$ and $\mathbf{e}_- = \mathbf{v}_T^- - \mathbf{v}_S^-$ denote the positive- and negative-branch errors, respectively. A positive-branch error can be offset by a compensating negative-branch error, since the composition leaves $\gamma \mathbf{e}_+ + (1 - \gamma) \mathbf{e}_-$ unchanged for any trade-off along that line. The composed loss can then reach zero while the student’s own conditional prediction still misses the reference-supplied appearance, with the negative-branch gap absorbing the error. We call this regime *Negative Branch Asymmetry* (NBA). Its signature is testable: because composed matching fixes only the composition at the training scale, not the underlying branches, the student can reproduce the teacher at γ_{train} yet drift as soon as guidance is recomposed at inference. Figure 1 illustrates this branch ambiguity and how supervising before composition removes it.

The remedy is to supervise before composition. Our primary objective, Positive-Direction Matching (PDM), constrains the positive prediction and the CFG conditional direction $\mathbf{v}^+ - \mathbf{v}^-$ separately. Zero loss now forces both branches into alignment, which removes the cancellation freedom while retaining the conditional signal CFG encodes, so supervision no longer depends on a single guidance scale. As a foil, we also study Independent Branch Matching (IBM), which matches the two branches directly; it removes the same ambiguity but weights the supervision differently.

We examine NBA in two settings that both create asymmetry through conditioning. For quantitative evidence, we study dense-to-sparse video control on Wan-VACE (Jiang et al., 2025), where the teacher observes dense per-frame guidance and the student receives only sparse keyframes: naive composed matching loses robustness as the inference guidance scale departs from training, while branch-aware objectives stay stable. As an image-domain qualitative case study, we return to the reference-conditioned examples above on FLUX (Labs, 2025): across guidance scales, naive matching degrades the reference-supplied appearance, whereas PDM more consistently preserves it. Both settings vary the conditioning available to teacher and student; whether architectural or capacity asymmetries induce NBA on their own is a question we leave open.

Our contributions are summarized as follows:

- We identify a hidden negative-branch alignment condition under which naive CFG-composed OPD is valid, and show that privileged conditioning violates it, leaving the composed objective under-identified at the branch level so that positive- and negative-branch errors can cancel. We term this regime *Negative Branch Asymmetry* (NBA).
- We introduce *Positive-Direction Matching* (PDM), a branch-aware OPD objective that supervises the positive prediction and the CFG conditional direction separately, restoring branch-level identifiability and robustness across inference guidance scales. We include Independent Branch Matching (IBM) as a branch-aware foil.
- We provide evidence in two conditioning-asymmetric settings: a quantitative dense-to-sparse video control benchmark and a qualitative reference-conditioned visual knowledge case study. Naive matching exhibits the guidance-scale sensitivity NBA predicts, while branch-aware supervision transfers knowledge more reliably.

2 RELATED WORK

2.1 DIFFUSION MODEL DISTILLATION

Although our goal is not few-step generation, prior diffusion distillation provides the closest context for understanding how teacher supervision and classifier-free guidance are typically transferred to a student. Most existing methods focus on reducing the number of denoising steps required at inference. Progressive Distillation recursively compresses multiple teacher steps into a single student step (Salimans & Ho, 2022). Consistency-based methods, including Consistency Models (Song et al., 2023; Kim et al., 2024) and Latent Consistency Models (Luo et al., 2023), train a model to produce consistent outputs across states along the same probability-flow trajectory, enabling one- or few-step generation. Distribution Matching Distillation (DMD) matches the student distribution to the teacher through their score functions (Yin et al., 2024b), while other approaches further combine diffusion supervision with adversarial training (Sauer et al., 2024; Yin et al., 2024a). Many of these methods also distill classifier-free guidance into the student. Rather than retaining separate positive and negative branches at inference, they use the CFG-composed teacher prediction as the distillation target, thereby baking the effect of guidance into a single student prediction (Luo et al., 2023; Yin et al., 2024b; Wu et al., 2026; Huang et al., 2025). This design improves sampling efficiency by avoiding two model evaluations for CFG and, more broadly, transfers the strongly guided teacher distribution into a compact few-step generator.

Our setting differs in both objective and purpose. We do not primarily seek to compress a multi-step guided teacher into a CFG-free few-step student. Instead, we study diffusion OPD in which the teacher and student each construct their own guided prediction from positive and negative branches. Under this formulation, matching the final guided predictions may fail to constrain the underlying positive branch when the two models are asymmetric on the negative branch.

2.2 ON-POLICY DISTILLATION FOR DIFFUSION MODELS

Recent work extends on-policy distillation to diffusion and flow models by querying the teacher on states sampled from the current student. D-OPSD performs on-policy self-distillation using privileged context, allowing a student to learn from a stronger teacher evaluated along the student’s own denoising trajectory (Jiang et al., 2026). Flow-OPD distills task-specialized flow-matching teachers into a unified student through on-policy sampling and dense per-step supervision (Fang et al., 2026).

DiffusionOPD formulates this process through transition-level KL divergence and shows that it reduces to prediction matching under standard diffusion transitions (Li et al., 2026b). Concurrently, DanceOPD treats generative capabilities as velocity fields and *collapses* operator-defined fields such as CFG into a single student prediction through on-policy field matching (Zhou et al., 2026); this is precisely the branch-collapsing design whose identifiability we analyze under negative-branch asymmetry.

Despite differences in derivation and optimization, these methods share the same central principle: the student generates the trajectory, while the teacher provides dense local supervision at student-visited states. Their studied configurations, however, either compress CFG into a single student prediction or instantiate teacher and student from the same base model with shared null-text conditioning, so their negative branches coincide by construction. They therefore do not examine the more general case where teacher–student conditioning produces a non-negligible negative-branch discrepancy.

The closest methods either collapse CFG into a single student prediction (guidance distillation, DanceOPD) or study symmetric teacher–student pairs whose negative branches coincide; neither examines faithful branch transfer under negative-branch asymmetry. Our work identifies this overlooked regime as NBA. We show that, under such asymmetry, directly matching the CFG-composed teacher and student predictions admits cross-branch cancellation and fails to uniquely constrain the positive branch. We further introduce branch-aware OPD objectives, with Positive–Direction Matching as the primary formulation, to resolve this ambiguity under negative-branch mismatch.

3 PRELIMINARIES

3.1 ON-POLICY DISTILLATION FOR DIFFUSION MODELS

Diffusion on-policy distillation (OPD) trains a student model using states sampled from its own evolving generation process (Jiang et al., 2026; Fang et al., 2026; Li et al., 2026b). Let

$$\mathbf{x}_t^S \sim p_\theta \quad (1)$$

denote a noisy state visited along a trajectory generated by the current student. At this state, the teacher and student define local denoising transitions conditioned on $\mathbf{c}_T$ and $\mathbf{c}_S$, respectively. The teacher may have access to privileged information unavailable or only partially available to the student, and therefore $\mathbf{c}_T$ and $\mathbf{c}_S$ need not be identical.

Existing diffusion OPD (Jiang et al., 2026; Fang et al., 2026; Li et al., 2026b) formulations minimize the discrepancy between the teacher and student transitions at student-visited states. Although they differ in their derivations and optimization procedures, their transition-level KL objectives reduce, up to a timestep-dependent weighting factor, to matching the corresponding model predictions:

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_{\mathbf{x}_t^S \sim p_{\theta, t}} \left[w(t) \left\| \mathbf{v}_T(\mathbf{x}_t^S, t, \mathbf{c}_T) - \mathbf{v}_S(\mathbf{x}_t^S, t, \mathbf{c}_S) \right\|_2^2 \right], \quad (2)$$

where $\mathbf{v}_T$ and $\mathbf{v}_S$ denote the teacher and student velocity predictions, and $w(t)$ absorbs the parameterization- and solver-dependent weighting. We use velocity prediction throughout the paper, while the same formulation applies to equivalent noise, score, or flow parameterizations.

3.2 DIFFUSION OPD WITH CLASSIFIER-FREE GUIDANCE

We next consider diffusion models that use classifier-free guidance (CFG), with guidance scale $\gamma > 1$, where the velocity used for denoising is

$$\tilde{\mathbf{v}} = \gamma \mathbf{v}^+ + (1 - \gamma) \mathbf{v}^-. \quad (3)$$

Here $\mathbf{v}^+$ and $\mathbf{v}^-$ denote the positive (conditional) and negative (unconditional) branch predictions. Define the positive- and negative-branch discrepancies as

$$\mathbf{e}_+ = \mathbf{v}_T^+ - \mathbf{v}_S^+, \quad \mathbf{e}_- = \mathbf{v}_T^- - \mathbf{v}_S^-. \quad (4)$$

Substituting the branch discrepancies into the CFG objective gives

$$\mathcal{L}_{\text{naive}} = \mathbb{E}_{\mathbf{x}_t^S \sim p_{\theta, t}} \left[w(t) \left\| \gamma \mathbf{e}_+ + (1 - \gamma) \mathbf{e}_- \right\|_2^2 \right]. \quad (5)$$

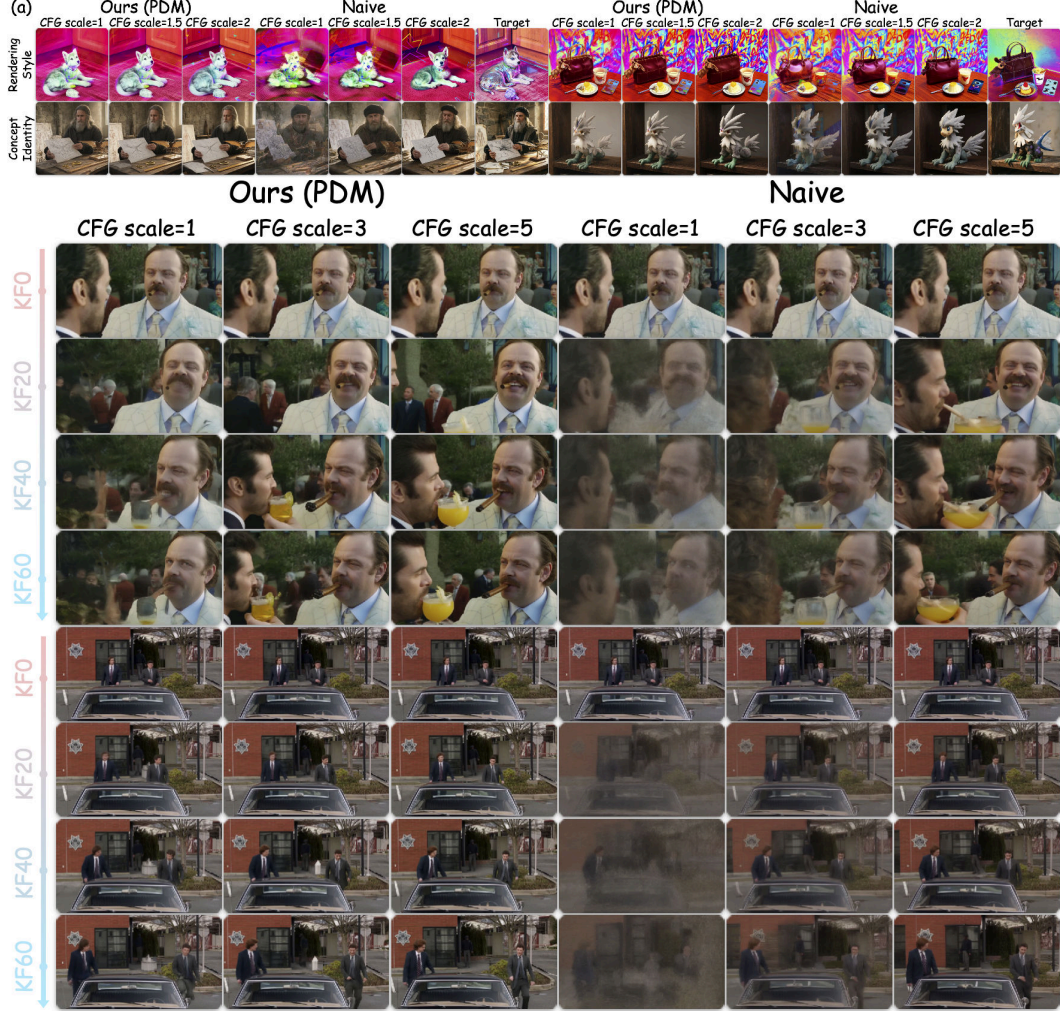


Figure 2: Qualitative manifestation of Negative Branch Asymmetry (NBA) under guidance-scale shifts. We compare PDM with naive CFG-based OPD on (a) visual knowledge distillation ($\gamma_{\text{train}} = 2$) and (b) dense-to-sparse video control ($\gamma_{\text{train}} = 5$). Naive OPD degrades when the inference guidance scale differs from the training configuration, exposing the branch-error cancellation permitted by guided-only matching. PDM remains more stable by separately matching the positive prediction and CFG conditional direction.

Existing OPD settings (Fang et al., 2026; Li et al., 2026b) are primarily studied in text-to-image distillation, where teacher and student are instantiated from the same base diffusion model (e.g., SD3.5 (Esser et al., 2024)) and share identical null-text conditioning in the negative branch. In this regime, the negative-branch discrepancy $\mathbf{e}_-$ is expected to be small, since teacher and student share negative-branch conditioning, causing CFG-based supervision to approximately reduce to positive-branch matching.

4 METHOD

4.1 NEGATIVE BRANCH ASYMMETRY

Once the teacher and student differ on the negative branch, Eq. 5 can no longer be approximated as positive-branch matching. Zero guided loss then only implies

$$\mathbf{e}_+ = \frac{\gamma - 1}{\gamma} \mathbf{e}_-, \quad (6)$$

which admits infinitely many non-zero solutions in the joint error space.

This implies that supervision on the CFG-composed direction is insufficient to uniquely constrain the positive-branch error, since errors in the negative branch can be arbitrarily traded off against positive-branch errors without affecting the objective value. We therefore refer to this phenomenon as **Negative Branch Asymmetry (NBA)**.

4.2 BRANCH-AWARE OPD

Based on this observation, we introduce Positive–Direction Matching (PDM), our primary branch-aware objective. PDM supervises the positive prediction together with the CFG conditional direction, thereby removing the cancellation ambiguity of guided-only matching. To contextualize this design, we additionally consider Independent Branch Matching (IBM), a straightforward baseline that directly matches the positive and negative branches separately.

For clarity, we use ℓ to denote the pointwise distillation loss evaluated at a single student-visited state $\mathbf{x}_t^S$ and timestep t . The corresponding OPD training objective is obtained by averaging this loss over the student-induced state distribution and timesteps:

$$\mathcal{L}_{\text{OPD}}[\ell] = \mathbb{E}_{\mathbf{x}_t^S \sim p_{\theta}, t} [w(t) \ell(\mathbf{x}_t^S, t)]. \quad (7)$$

All branch-aware objectives below specify different choices of the pointwise loss ℓ , while sharing the same on-policy sampling and timestep weighting in Eq. 7.

Positive–Direction Matching (PDM). Our primary branch-aware objective is Positive–Direction Matching (PDM), which supervises the positive prediction together with the CFG conditional direction.

For a model $M \in \{T, S\}$, we define its conditional direction as

$$\mathbf{d}_M = \mathbf{v}_M^+ - \mathbf{v}_M^-. \quad (8)$$

The pair $(\mathbf{v}_M^+, \mathbf{d}_M)$ uniquely determines the underlying positive and negative predictions, and therefore the CFG-composed prediction at any guidance scale. The corresponding teacher–student direction discrepancy is

$$\mathbf{d}_T - \mathbf{d}_S = \mathbf{e}_+ - \mathbf{e}_-. \quad (9)$$

We define the PDM loss as

$$\ell_{\text{PDM}} = \|\mathbf{e}_+\|_2^2 + \lambda \|\mathbf{d}_T - \mathbf{d}_S\|_2^2 = \|\mathbf{e}_+\|_2^2 + \lambda \|\mathbf{e}_+ - \mathbf{e}_-\|_2^2, \quad (10)$$

where $\lambda > 0$ controls the strength of conditional-direction matching.

The first term anchors the positive prediction, while the second preserves the CFG conditional direction. For $\lambda > 0$, zero PDM loss requires $\mathbf{e}_+ = \mathbf{e}_- = 0$, thereby eliminating the cancellation ambiguity of naive guided matching.

Alternative baseline: Independent Branch Matching (IBM). For comparison, we consider Independent Branch Matching (IBM), a direct branch-aware baseline that matches the positive and negative predictions separately:

$$\ell_{\text{IBM}} = \gamma^2 \|\mathbf{e}_+\|_2^2 + (\gamma - 1)^2 \|\mathbf{e}_-\|_2^2. \quad (11)$$

For the standard guided regime $\gamma > 1$, IBM has the same unique branch-level zero-loss solution as PDM. The two objectives remove the same cancellation ambiguity but parameterize and weight the supervision differently; we compare their behavior empirically in Table 1.

Efficient partial-trajectory supervision. Applying OPD supervision to every state along a full denoising trajectory requires repeated teacher evaluations and substantially increases training cost. We empirically find that such full-trajectory supervision is unnecessary: using only a short prefix of student-visited states can preserve distillation quality while greatly improving training efficiency. A related observation is independently reported in concurrent work (Zhou et al., 2026). Motivated by this observation, we introduce partial-trajectory supervision.

Let $\{\mathbf{x}_{t_k}^S\}_{k=0}^{N-1}$ denote the states visited by the student in rollout order. We accumulate the distillation objective over only the first K states:

$$\mathcal{L}_{\text{OPD}}^{(K)} = \mathbb{E}_{\{\mathbf{x}_{t_k}^S\}_{k=0}^{N-1} \sim p_\theta} \left[\frac{1}{K} \sum_{k=0}^{K-1} w(t_k) \ell(\mathbf{x}_{t_k}^S, t_k) \right], \quad (12)$$

where ℓ can be any pointwise OPD objective defined above. The student still completes the standard denoising rollout, but teacher supervision is computed only for these K states. The hyperparameter K therefore controls the trade-off between supervision coverage and training efficiency, with $K = N$ recovering full-trajectory supervision. As shown in Section 5.3.2, a moderate horizon substantially reduces training cost while maintaining strong distillation performance.

4.3 TASK INSTANTIATIONS

4.3.1 VISUAL KNOWLEDGE DISTILLATION

We consider D-OPSD-based (Jiang et al., 2026) visual knowledge distillation using a multi-step FLUX (Labs, 2025) model with CFG. The teacher observes a reference image $\mathbf{r}$ containing visual knowledge that is not fully specified by the text prompt, while the student is conditioned only on the text prompt $\mathbf{y}$. The reference-supplied knowledge may correspond to a transferable rendering style and the visual identity of a particular concept (Wang et al., 2026a). Their positive conditions are therefore

$$\mathbf{c}_T^+ = (\mathbf{y}, \mathbf{r}), \quad \mathbf{c}_S^+ = \mathbf{y}. \quad (13)$$

Importantly, the reference image remains available to the teacher when constructing its negative prediction:

$$\mathbf{c}_T^- = (\emptyset, \mathbf{r}), \quad \mathbf{c}_S^- = \emptyset, \quad (14)$$

where $\emptyset$ denotes the null-text condition. Consequently, the teacher and student are asymmetric not only on the positive branch but also on the negative branch, making this setting a direct instance of NBA.

4.3.2 DENSE-TO-SPARSE VIDEO CONTROL

We next study video control distillation in the dense-to-sparse setting using the Wan-VACE (Jiang et al., 2025) video generation model. The teacher receives a dense control sequence $\mathbf{p}_{\text{dense}}$, while the student observes only a sparse subset $\mathbf{p}_{\text{sparse}}$. Their conditions are

$$\begin{aligned} \mathbf{c}_T^+ &= (\mathbf{y}, \mathbf{p}_{\text{dense}}), & \mathbf{c}_S^+ &= (\mathbf{y}, \mathbf{p}_{\text{sparse}}), \\ \mathbf{c}_T^- &= (\emptyset, \mathbf{p}_{\text{dense}}), & \mathbf{c}_S^- &= (\emptyset, \mathbf{p}_{\text{sparse}}). \end{aligned} \quad (15)$$

The dense and sparse controls therefore induce different negative predictions even when both branches use the same null-text condition.

Table 1: Main comparison under joint multi-modal training. Each model is jointly trained on pose, depth, and scribble controls and evaluated separately on the three modalities. **Bold** and underlined numbers denote the best and second-best results, excluding Teacher.

Pose-Control		MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓	
Method		All	Key	All	Key	All	Key	FID	FVD
Teacher		3.03	2.82	96.02	96.46	91.15	92.23	14.13	56.01
Student		5.92	4.10	86.78	91.89	69.97	82.53	15.81	81.41
SFT		5.02	3.31	90.36	94.77	76.72	88.55	14.11	63.90
Off-policy		5.14	3.44	89.78	94.09	75.46	87.48	14.49	68.50
OPD (Naive)		4.43	3.14	91.84	95.25	80.28	89.35	<u>13.49</u>	<u>65.12</u>
OPD (IBM)		<u>4.20</u>	<u>2.88</u>	<u>92.93</u>	<u>96.11</u>	<u>82.27</u>	<u>90.95</u>	13.26	65.72
OPD (PDM)		4.13	2.83	92.98	96.12	82.48	91.03	13.52	62.77
Depth-Control		CORR ↑		RMSE ↓		SI-RMSE ↓		Quality ↓	
Method		All	Key	All	Key	All	Key	FID	FVD
Teacher		96.98	97.58	7.50	6.36	6.19	5.23	11.02	36.89
Student		83.21	91.27	16.87	11.13	14.36	9.37	15.03	71.33
SFT		88.39	96.17	13.83	7.80	11.96	6.52	12.87	58.13
Off-policy		88.25	96.01	13.98	8.00	12.10	6.64	12.66	<u>59.35</u>
OPD (Naive)		90.58	96.56	12.39	7.47	10.69	6.16	12.08	72.20
OPD (IBM)		<u>91.14</u>	<u>96.91</u>	<u>12.03</u>	<u>7.11</u>	<u>10.37</u>	<u>5.85</u>	<u>11.76</u>	60.38
OPD (PDM)		91.24	97.13	11.95	6.91	10.31	5.68	11.64	60.60
Scribble-Control		Recall ↑		Precision ↑		F1 ↑		Quality ↓	
Method		All	Key	All	Key	All	Key	FID	FVD
Teacher		93.26	94.40	94.26	95.27	93.73	94.81	9.56	24.22
Student		64.20	82.93	65.50	83.78	64.70	83.24	12.29	69.13
SFT		67.23	89.03	69.52	90.66	68.27	89.79	10.69	50.56
Off-policy		70.98	91.55	72.15	91.73	71.49	91.60	11.60	52.90
OPD (Naive)		74.78	90.77	75.38	91.34	75.00	90.99	10.53	59.95
OPD (IBM)		<u>75.92</u>	<u>92.23</u>	76.47	<u>92.70</u>	<u>76.13</u>	<u>92.43</u>	10.12	<u>52.67</u>
OPD (PDM)		76.42	93.03	<u>76.13</u>	92.86	76.22	93.03	<u>10.34</u>	50.53

5 EXPERIMENTS

We evaluate whether naive CFG-based OPD suffers from instability under NBA, and whether branch-aware objectives improve both optimization stability and generalization across guidance and conditioning variations.

5.1 EXPERIMENTAL SETUP

Visual knowledge distillation. We evaluate the visual knowledge distillation setting introduced in Section 4.3.1 on held-out prompts spanning rendering style and concept identity. We use this experiment as a qualitative case study of NBA: all students are trained at a fixed guidance scale ($\gamma_{\text{train}} = 2$) and evaluated across different inference guidance scales, revealing whether guided-only matching recovers transferable branch-level predictions or merely fits a particular CFG composition. We compare naive CFG-based OPD with PDM under the same training setup. Additional details are provided in Appendix 7.1.1.

Dense-to-sparse video control. We construct a dense-to-sparse video control benchmark from OpenHumanVideo (Li et al., 2025), containing 600 test clips across diverse subjects, motions, and interaction scenarios. We evaluate pose, depth, and scribble control under the dense-to-4-keyframe setting, with all methods evaluated on full video rollouts against the ground-truth videos.

Table 2: Guidance-scale generalization in the pose-only ablation setting. All models are trained only on pose control at a fixed guidance scale $\gamma_{\text{train}} = 5$ and evaluated at different inference guidance scales.

Method	MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓	
	All	Key	All	Key	All	Key	FID	FVD
Naive ($\gamma = 5$)	4.62	3.22	91.35	95.01	79.39	89.54	14.05	67.44
Naive ($\gamma = 3$)	4.94	3.31	90.23	94.53	77.92	88.22	22.66	134.09
Naive ($\gamma = 1$)	8.98	6.23	80.08	87.42	69.14	81.57	78.20	507.70
IBM ($\gamma = 5$)	4.28	2.91	92.67	95.95	81.69	91.03	12.99	53.83
IBM ($\gamma = 3$)	4.42	3.04	92.09	95.53	81.41	90.81	13.69	56.85
IBM ($\gamma = 1$)	4.91	3.43	90.92	94.69	80.66	89.98	18.00	71.17
PDM ($\gamma = 5$)	4.27	2.75	92.71	96.38	82.12	91.66	13.83	58.65
PDM ($\gamma = 3$)	4.17	2.73	92.81	96.61	82.27	91.81	13.46	55.85
PDM ($\gamma = 1$)	4.48	2.90	92.14	95.64	81.72	91.15	15.25	60.97

Unless otherwise specified, we use $\lambda = 1$ for PDM and supervise the first $K = 8$ student-visited transitions. For the main comparison, each method is jointly trained on pose, depth, and scribble controls and evaluated separately on each modality. The guidance-scale and hyperparameter ablations instead use pose-only models to isolate the factor being studied and reduce repeated training cost. Results obtained under the joint-modality and pose-only protocols are therefore not directly comparable. Complete benchmark, evaluation, and implementation details are provided in Appendix 7.1.

5.2 MAIN RESULTS

5.2.1 VISUAL KNOWLEDGE DISTILLATION

As shown in Figure 2, naive CFG-based OPD fails to transfer reference-supplied visual knowledge consistently: it weakens the target attribute and drifts further as the inference guidance scale moves away from the training configuration. PDM preserves the defining reference attribute more consistently across guidance scales while still following the evaluation prompt. This pattern is consistent with the cancellation ambiguity predicted by NBA—fitting one CFG composition need not recover the underlying reference-conditioned prediction. Because the evidence here is qualitative, we treat it as an illustrative case study rather than an aggregate estimate of performance.

5.2.2 DENSE-TO-SPARSE VIDEO CONTROL DISTILLATION

Dense-to-Sparse Distillation under Training Guidance We first evaluate all methods using joint training over pose, depth, and scribble controls. Table 1 reports results under the default guidance configuration, where the training and inference guidance scales are both set to $\gamma = 5$. Naive CFG-based OPD underperforms branch-aware objectives on control-fidelity metrics across all three modalities, despite using the same CFG configuration during training and inference. Our identifiability analysis does not by itself predict this matched-scale gap: at a fixed guidance scale the composed objective is a well-defined target. We attribute the gap to a proxy mismatch, since a lower composed-velocity loss need not translate into higher control fidelity; the sharper, identifiability-driven failure that NBA predicts emerges when the inference guidance scale departs from training (Table 2).

Empirically, PDM delivers the most consistent overall improvements across control modalities, while IBM also outperforms naive guided matching on most control-fidelity metrics.

Generalization across Guidance Scales The previous experiments evaluate all methods under the same guidance configuration used during training. However, NBA predicts that matching only the CFG-composed prediction does not identify the underlying positive and negative branches, and therefore may fail when the guidance composition changes.

Table 2 evaluates the generalization of different objectives across inference guidance scales. Naive OPD exhibits severe degradation when the guidance scale deviates from the training configuration,



Figure 3: Qualitative comparison on dense-to-sparse pose control distillation under the default guidance scale $\gamma = 5$. Compared with SFT, off-policy distillation, and naive CFG-based OPD, Positive-Direction Matching (PDM) produces more faithful motion trajectories and higher visual quality under sparse control supervision.

Table 3: Effect of the conditional-direction weight λ in the pose-only ablation setting. **Bold** and underlined numbers denote the best and second-best results, respectively.

Method	MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓	
	All	Key	All	Key	All	Key	FID	FVD
PDM ($\lambda = 0$)	4.24	2.81	92.71	96.24	81.80	91.05	13.97	59.72
PDM ($\lambda = 1$)	<u>4.27</u>	2.75	92.71	96.38	82.12	91.66	13.83	58.65
PDM ($\lambda = 3$)	4.36	2.95	<u>92.08</u>	95.77	81.25	90.39	13.33	61.79
PDM ($\lambda = 5$)	4.40	3.00	<u>92.27</u>	95.96	80.89	90.75	<u>12.97</u>	56.85
PDM ($\lambda = 10$)	4.49	3.07	91.60	95.35	80.21	89.93	12.96	<u>56.91</u>

with the performance dropping substantially at $\gamma = 1$. In contrast, branch-aware objectives remain robust across different guidance scales, confirming that explicitly constraining branch-level information provides more transferable supervision.

Both branch-aware objectives substantially improve guidance-scale robustness over naive OPD. In this pose-control setting, PDM exhibits the most stable performance across inference guidance scales.

5.3 ABLATION STUDIES

5.3.1 EFFECT OF CONDITIONAL-DIRECTION WEIGHT

We study the effect of λ in PDM, which controls the strength of conditional-direction preservation. When $\lambda = 0$, PDM reduces to positive-only matching and does not explicitly constrain the CFG conditional direction.

Table 4: Effect of the OPD supervision horizon K in the pose-only ablation setting. Wall-clock training time is reported in seconds per optimization step. **Bold** and underlined numbers denote the best and second-best results, respectively.

Method	MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓		Time ↓
	All	Key	All	Key	All	Key	FID	FVD	s/step
PDM ($K = 1$)	4.33	2.98	92.53	95.68	81.19	90.57	<u>13.70</u>	54.41	~23
PDM ($K = 2$)	4.45	3.11	92.18	95.57	80.78	90.51	13.89	60.63	~38
PDM ($K = 4$)	4.27	<u>2.86</u>	<u>92.59</u>	<u>95.98</u>	<u>81.71</u>	<u>91.13</u>	13.58	<u>57.08</u>	~69
PDM ($K = 8$)	4.27	2.75	92.71	96.38	82.12	91.66	13.83	58.65	~132
PDM ($K = 50$)	4.65	3.14	91.34	95.25	80.81	90.09	13.91	61.95	~790

As shown in Table 3, $\lambda = 1$ provides the best overall balance between control fidelity and generation quality. In particular, it achieves the lowest keyframe MPJPE and the highest PCK@0.1 scores, while also improving FVD over positive-only matching. Increasing λ beyond 1 improves some distributional-quality metrics, with $\lambda = 5$ and $\lambda = 10$ producing lower FID and FVD, but consistently weakens control accuracy. This suggests that overemphasizing direction matching introduces a trade-off between visual quality and adherence to the sparse control signal.

Because control fidelity is the primary objective in our dense-to-sparse setting, while generation quality must remain competitive, we use $\lambda = 1$ as the default value in all main experiments.

5.3.2 EFFECT OF ON-POLICY SUPERVISION HORIZON

We empirically study how the number of supervised student-visited states affects training efficiency and distillation performance. The student generates a full denoising trajectory, while only the first K visited states contribute to the distillation objective.

Table 4 reveals a non-monotonic empirical trade-off. Increasing K substantially raises the training cost, from 23 seconds per step at $K = 1$ to 790 seconds per step under full-trajectory supervision ($K = 50$). Among the evaluated settings, $K = 8$ achieves the strongest control fidelity, including the lowest keyframe MPJPE and the highest PCK scores. Although $K = 1$ and $K = 4$ achieve better individual distributional-quality metrics at lower cost, they provide weaker adherence to the sparse control signals.

These results should be interpreted as an empirical observation under our training setup rather than a theoretical characterization of the optimal supervision horizon. We use $K = 8$ as the default because it provides the best observed control fidelity while remaining substantially more efficient than full-trajectory supervision.

5.4 QUALITATIVE RESULTS

Figure 2 qualitatively illustrates NBA in both visual knowledge distillation and dense-to-sparse video control. When the inference guidance scale differs from the training configuration, naive CFG-based OPD exhibits substantial degradation, whereas PDM remains stable across guidance scales.

Figure 3 further compares PDM with SFT, off-policy distillation, and naive OPD under the default guidance scale. PDM more faithfully follows the sparse pose controls while producing coherent intermediate frames, showing that its improvements are not limited to guidance-scale shifts. Additional qualitative results are provided in Appendix 7.2.

6 CONCLUSIONS AND LIMITATIONS

We identify Negative Branch Asymmetry (NBA), a failure mode of CFG-based OPD in which teacher–student discrepancies on the positive and negative branches can cancel under guided prediction matching. To address this ambiguity, we introduce Positive–Direction Matching (PDM), which supervises the positive prediction and CFG conditional direction separately. Experiments on

visual knowledge distillation and dense-to-sparse video control show that PDM improves distillation performance and remains more stable across inference guidance scales.

Our study is limited to linear CFG and two model families, while the visual knowledge results are primarily qualitative. Moreover, PDM and IBM share the same branch-level zero-loss solution under non-degenerate weights, so PDM’s advantage is currently empirical rather than theoretical. Future work may extend NBA analysis to other guidance mechanisms and develop more efficient on-policy supervision.

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7 APPENDIX

7.1 ADDITIONAL EXPERIMENTAL DETAILS

7.1.1 VISUAL KNOWLEDGE DISTILLATION

Data and task construction. We construct a small-scale visual knowledge distillation setting with 24 reference–prompt pairs (Wang et al., 2026a;b). Each pair specifies one visual-knowledge target and contains one reference image that exposes the target appearance to the teacher. We treat the examples collectively as visual-knowledge targets rather than assigning them to disjoint style, concept, or world-knowledge categories. To increase the number of stochastic training observations without introducing additional targets, we repeat each pair 16 times in the training manifest, yielding 384 sampling records. Different visits still use independently sampled initial noise and denoising states. For qualitative evaluation, we construct held-out prompts around the same key visual knowledge while changing the surrounding context, composition, or interactions. The resulting evaluation tests whether the text-only student internalizes the reference-supplied visual knowledge instead of reproducing a particular training scene.

Model and conditioning. We use the multi-step FLUX.2-klein-4B-base model (Labs, 2025) at 512×512 resolution. Each reference image is converted to RGB, bicubically resized to cover the target canvas, center-cropped, and normalized to $[-1, 1]$ before VAE encoding. The student follows the original text-to-image interface and receives only the prompt. The teacher uses the same diffusion backbone in its image-editing interface: it receives the student-visited noisy latent together with the encoded reference-image latents, and its text condition appends the instruction “Make the main subject (facial/IP details), background, and overall composition consistent with the reference image.” For the teacher’s negative prediction, we replace the text with the null prompt while retaining the reference-image latents. The student’s negative prediction uses the same null prompt without reference latents. The reference therefore induces the negative-branch asymmetry analyzed in Section 4.3.1.

Parameterization and teacher update. We freeze the VAE, text encoder, and base diffusion-transformer weights and optimize LoRA adapters with rank 64 and scaling parameter $\alpha = 128$. The adapters are applied to the query, key, value, and output projections of the self- and context-attention modules, together with the corresponding feed-forward projections in each transformer block. We maintain separate student and teacher adapters initialized from the same weights. After each student update, we update the teacher adapter with exponential moving average decay 0.9999. Teacher predictions are evaluated without gradients.

On-policy objectives. For each batch, the student initializes a trajectory from Gaussian noise under the text-only condition. We use the native FLUX.2-klein scheduler with a 30-step denoising grid and supervise the first $K = 8$ student-visited transitions. At every supervised state, the teacher and student construct their own positive and negative predictions, and both rollouts use the CFG-composed velocity associated with the training guidance scale $\gamma_{\text{train}} = 2$. We implement naive OPD by matching the teacher and student after CFG composition. PDM instead matches the teacher’s positive prediction and the CFG conditional direction separately, with $\lambda = 1$. All compared objectives use the same base model, reference–prompt pairs, student rollout states, and optimization budget.

Optimization. We train with AdamW using learning rate 10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, zero weight decay, and gradient-norm clipping at 1.0. Training uses eight processes with a per-process batch size of 2 and no gradient accumulation, giving a global batch size of 16. We use bfloat16 mixed precision for the trainable model and VAE, gradient checkpointing, and DeepSpeed ZeRO-2. We allow at most 3001 optimization updates and select the checkpoint for qualitative evaluation after the corresponding training loss has converged, using the same selection rule for every objective.

Inference and qualitative evaluation. At inference, we discard the teacher adapter and reference image; the student receives only the held-out text prompt. We generate 512×512 images with 40 denoising steps and evaluate guidance scales $\gamma \in \{1, 1.5, 2\}$. For a given evaluation prompt, all objectives and guidance scales use the same initial noise, isolating changes caused by the training

Bucket	Motion score	#Test	#Train
diff1.5-2.0	[1.5, 2.0)	100	2,194
diff2.0-2.5	[2.0, 2.5)	100	6,355
diff2.5-3.5	[2.5, 3.5)	100	8,992
diff3.5-5.0	[3.5, 5.0)	100	4,851
diff5.0-7.0	[5.0, 7.0)	100	1,319
diff7.0+	[7.0, ∞)	100	364
Total		600	$\approx 24K$

Table 5: Difficulty-stratified benchmark. The test set draws 100 clips from each of six motion-difficulty buckets (600 total); motion score is the mean frame-to-frame pixel difference of the rendered DWPose skeleton.

objective and CFG composition. We assess the resulting images qualitatively by inspecting whether they preserve the defining visual knowledge while following the changed context of the evaluation prompt and maintaining generation quality. This experiment is an illustrative qualitative case study; we do not interpret it as an aggregate estimate of performance.

7.1.2 DENSE-TO-SPARSE VIDEO BENCHMARK

Our dense-to-sparse video benchmark is derived from OpenHumanVid (OHV) (Li et al., 2025). Starting from the 12-part OHV release ($\approx 91K$ clips with DWPose annotations), we first filter clips according to minimum length and motion magnitude. We then split videos into non-overlapping 81-frame clips at 832×480 resolution while preserving pixel alignment between RGB and control streams.

Difficulty stratification. To reduce evaluation bias toward easier samples (Li et al., 2026a), we compute a motion score based on frame-to-frame differences of rendered DWPose skeletons. The clips are divided into six difficulty buckets according to this score, and we randomly sample 100 clips from each bucket to form the 600-clip test set. The remaining clips are used for training and validation. Please refer to Table 5 for more details.

7.1.3 SPARSE KEYFRAME CONTROL

All video experiments target the dense-to-sparse setting: the teacher receives the control signal at *every* frame, while the student receives it only at a sparse set of pixel-space keyframes. Following the masked video-to-video (MV2V) interface of the VACE (Jiang et al., 2025) base model, the student input frame stack F and mask M (over 81 frames at 480×832 , in the $[-1, 1]$ range) are built as

- $F[0]$ = the reference image (appearance anchor), with $M[0] = 0$ (given, not generated);
- $F[k]$ = the control frame at each pose keyframe $k \in \{20, 40, 60\}$;
- all remaining frames are blank (-1), with $M[\cdot] = 1$ (to be synthesized).

We use the uniform layout `pixel_kf` = $\{0, 20, 40, 60\}$ throughout: index 0 is reserved for the reference image, so the model is conditioned on pose at three interior keyframes out of 81 frames. Keyframes are indexed in *pixel* space; under the VACE VAE ($4 \times$ temporal, $8 \times$ spatial compression) this maps non-linearly to latent frames, so all masks are constructed in pixel space and then compressed. This same sparse construction is used both inside the on-policy rollout during training and at inference; the teacher instead receives the dense per-frame control.

7.1.4 EVALUATION METRICS

All methods are evaluated independently on the 600-clip test set. Scores are computed per clip and averaged over the dataset.

Control fidelity (per modality). For each control modality we *re-extract* the control signal from the generated video with the corresponding annotator and compare it to the control that was fed in.

<i>Base model & adapter</i>	
Base model	Wan2.1-VACE-1.3B (frozen)
Adapter	LoRA, rank 32, $\alpha = 32$
LoRA target	self/cross-attn $\{q, k, v, o\}$, ffn $\{0, 2\}$ per VACE block
Precision	bf16 autocast, gradient checkpointing
<i>Optimization</i>	
Optimizer	AdamW, $\beta = (0.9, 0.999)$, wd = 0, $\epsilon = 10^{-8}$
Learning rate	1×10^{-4} (constant)
Training steps	2000
Devices / batch	8 GPUs (DDP), per-GPU batch 1
<i>Diffusion & rollout</i>	
Solver / schedule	UniPC, 50 steps, <code>shift</code> = 16
OPD rollout K	8 active steps ("8-of-50")
Guidance scale γ	5.0
<i>Data</i>	
Resolution / length	480×832 , 81 frames
Keyframes	pixel $\{0, 20, 40, 60\}$ (0 = reference)
Train / val / test	$\approx 24\text{K} / 24 / 600$ clips

Table 6: Training and inference hyperparameters, shared across all distillation objectives.

- **Pose.** We run DWPose (Yang et al., 2023) on the generated video and use the 18 body joints, in normalized $[0, 1]$ image coordinates. A joint is counted only when both the GT and predicted confidence exceed 0.3; frames with fewer than 3 valid joints are skipped. We report *MPJPE* (mean per-joint ℓ_2 error in normalized coordinates) and *PCK@ τ* for $\tau \in \{0.2, 0.1\}$, where a joint is correct if its error is below $\tau \cdot s$ and s is the per-frame bounding-box size $s = \max(\Delta x, \Delta y)$ of the GT skeleton. Metrics are reported over *all* frames and over the *keyframe* subset separately.
- **Depth.** We re-extract depth from the generated frames with DPT-Hybrid (MiDaS) (Ranftl et al., 2020) and compare against the dense depth control (same annotator). Since these are *relative* depths, we remove the scale/shift ambiguity with a closed-form least-squares affine alignment $a \cdot \hat{d} + b \approx d$ and report the affine-invariant *SI-RMSE* (primary), the raw RMSE, and the Pearson correlation.
- **Scribble.** We re-extract neural line-art from the generated frames with the VACE scribble annotator and compare against the dense line-art control. Line pixels are binarized as deviations from the per-frame median ($> 40/255$), and we report a tolerant boundary *F1* (precision/recall) with a dilation tolerance of 2 pixels.

Distributional quality. Independent of control adherence, we measure perceptual and temporal realism against the GT distribution: *FID* (clean-fid, InceptionV3 features, 8 frames per clip) captures per-frame appearance, and *FVD* (I3D pretrained on Kinetics-400, 16 frames linspace-sampled per clip) captures spatio-temporal consistency.

7.1.5 IMPLEMENTATION AND HYPERPARAMETERS

All methods fine-tune the same Wan2.1-VACE-1.3B (Jiang et al., 2025) base model with LoRA (Hu et al., 2022) adapters (the base weights are frozen), so differences are attributable to the training objective alone. LoRA (rank 32, $\alpha = 32$) is applied to the query/key/value/output projections of both self- and cross-attention and the two feed-forward linears within every VACE block. We train with AdamW (lr = 10^{-4} , $\beta = (0.9, 0.999)$, no weight decay) for 2000 steps on 8 GPUs (DDP, per-GPU batch 1), under bf16 mixed precision with gradient checkpointing. Inference and the on-policy rollout use the UniPC solver (Zhao et al., 2023) on a 50-step schedule (`shift`= 16). All models are distilled and deployed at classifier-free guidance scale $\gamma = 5$. Videos are 81 frames at 480×832 . Table 6 lists the full configuration.

Off-policy distillation baseline. The off-policy baseline uses the same teacher–student prediction-matching loss as OPD but evaluates it on standard data-derived diffusion states. Given a ground-

truth video latent $\mathbf{x}_0$, we sample t and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and construct $\mathbf{x}_t^{\text{off}} = \alpha_t \mathbf{x}_0 + \sigma_t \epsilon$. While SFT matches the student prediction at this state to the ground-truth velocity target, the off-policy baseline replaces that target with the teacher prediction under dense control. The student is evaluated at the same state under sparse control. It therefore differs from OPD only in the state distribution, using noised ground-truth latents instead of student-visited states.

7.2 MORE QUALITATIVE RESULTS

Figures 4 and 5 provide additional qualitative evidence of NBA under guidance-scale shifts. In both concept identity distillation and dense-to-sparse video control, naive CFG-based OPD degrades when the inference guidance scale differs from the training configuration. In contrast, PDM more consistently preserves the transferred visual knowledge and controlled video structure across guidance scales.

Figures 6 and 7 extend the qualitative comparison to depth and scribble control under the default guidance scale. Across both modalities, PDM follows the sparse structural conditions more faithfully while maintaining coherent visual content. These results show that the benefit of branch-aware supervision is consistent across different forms of conditioning asymmetry.



Figure 4: Qualitative comparison on concept identity distillation across inference CFG scales. The rightmost column shows the target appearance to be internalized by the text-only student. PDM (Ours) consistently preserves the defining visual attributes of each target across guidance scales, whereas naive OPD exhibits identity drift and degraded generation quality, particularly when the inference guidance scale changes. These results illustrate the guidance-scale sensitivity induced by NBA.

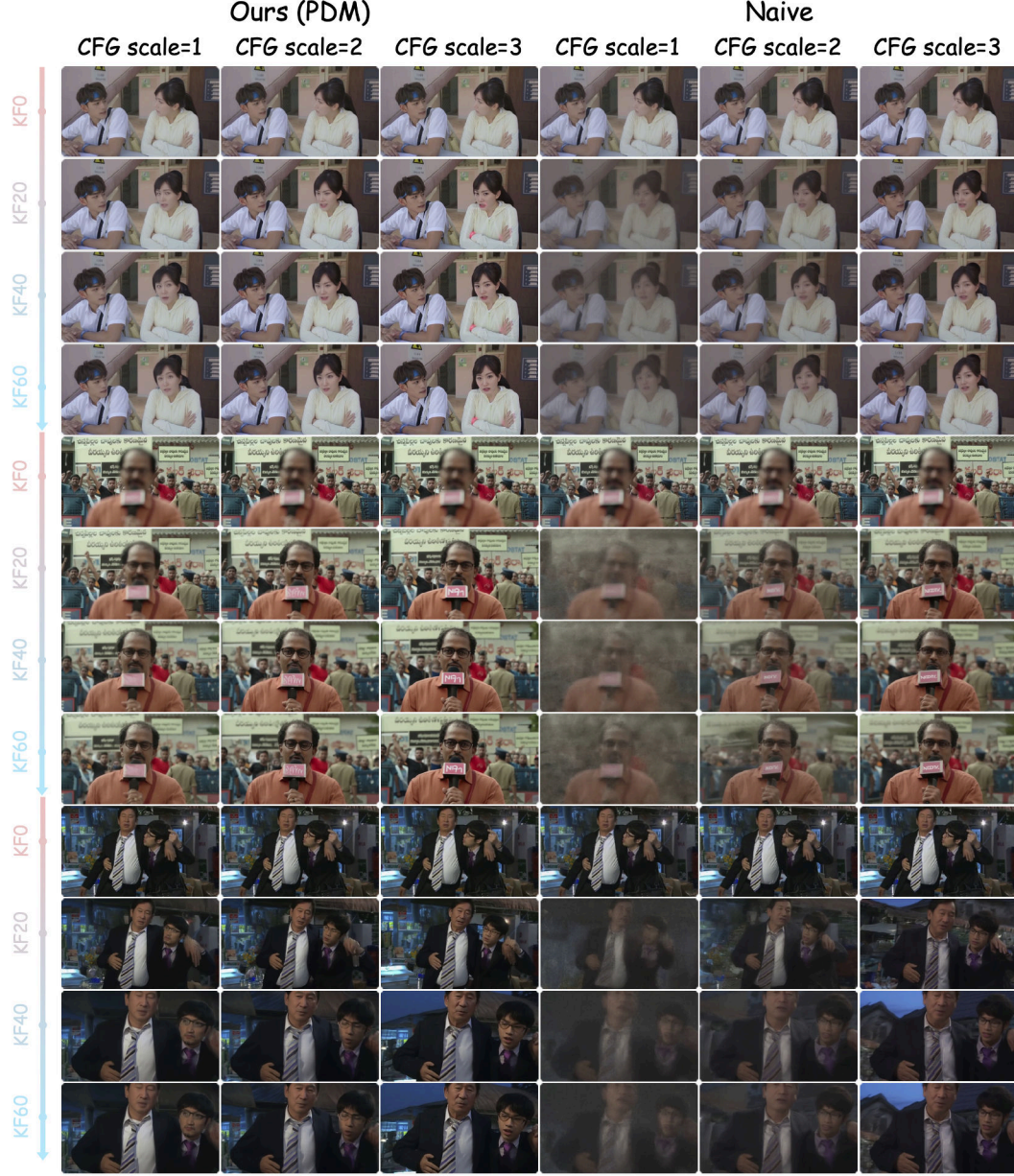


Figure 5: Additional qualitative visualization of NBA in dense-to-sparse video control. Models are trained at $\gamma_{\text{train}} = 5$ and evaluated at lower inference guidance scales. Naive CFG-based OPD exhibits severe blurring and structural degradation as γ_{test} moves away from the training configuration, revealing its sensitivity to a specific CFG composition. PDM consistently preserves subject appearance and video structure across guidance scales by separately matching the positive prediction and CFG conditional direction.



Figure 6: Qualitative comparison on dense-to-sparse depth control distillation under the default guidance scale $\gamma = 5$. The teacher receives dense depth guidance, while the student is conditioned only on sparse keyframes. PDM achieves more consistent structure and visual quality compared with other distillation objectives.



Figure 7: Qualitative results on dense-to-sparse scribble control distillation with $\gamma = 5$. Compared with other distillation objectives, Positive-Direction Matching better preserves the scribble-guided structure and generates more consistent visual results under sparse control supervision.